

Mathematical Derivations Explained from First Principles

Beyond Grounding: Bias-Aware Selective LLM Question Answering over Biased Records

Supplementary companion for ICTAI 2026. This version is intentionally written for readers who may have no background in probability, statistics, machine learning, or Bayesian methods. Every technical term is introduced before it is needed, and every numbered equation is built from simpler ideas.

What problem are these equations trying to solve?

Suppose a database contains a number such as a product’s carbon intensity. The number written in the database is not automatically the same thing as the unknown value we would ideally like to know. A measurement can contain ordinary random error, and an entire group of records can also share a systematic shift. For example, one supplier group might report values that are consistently higher or lower because of a measurement or reporting process.

The method in the paper tries to reason about three things at once:

- the values we actually observe in the records;
- the cleaner values that may have existed before measurement or reporting distortion; and
- how confident the system should be before it gives an answer instead of refusing to answer.

There is an unavoidable difficulty. If a recorded value is 80, we cannot tell from that number alone whether the clean value was 80 with no systematic shift, 75 with a shift of 5, or 70 with a shift of 10. In symbols,

$$80 = 80 + 0 = 75 + 5 = 70 + 10.$$

All three explanations fit the same recorded number. This is why the method needs some outside information about the expected difference between groups. The paper calls that outside reference an *anchor*. The anchor does not magically create truth; it supplies information that the biased records themselves do not contain.

In plain English. The equations are not trying to “discover truth from nothing.” They combine the observed records, a model of how errors may occur, and an external anchor. The final answer is therefore conditional on those assumptions and inputs.

Small mathematical toolbox

This section introduces the minimum vocabulary needed later. A reader who already knows these ideas can move directly to Equation (1).

A symbol can stand for an unknown number

A letter such as X can stand for a number whose exact value is not currently known. If the clean carbon intensity of a product is unknown, we may call that clean value X . If the database records 72.4, we may call the recorded value $W = 72.4$.

Uppercase letters are often used for quantities that can vary across repeated situations; lowercase letters often denote one particular value. This convention is useful but not essential. The document always explains the meaning in words.

Mean, variance, and standard deviation

The **mean** is an average. For numbers $x_1, \dots, x_n$,

$$\bar{x} = \frac{x_1 + \dots + x_n}{n}.$$

The bar over x means “average.”

The **variance** measures spread by averaging squared distances from the mean. The **standard deviation** is the square root of the variance. The reason standard deviation is convenient is that it returns to the original measurement unit. If carbon intensity is measured in kg CO₂/kWh, its variance uses squared units, while its standard deviation again uses kg CO₂/kWh.

We write

$$\text{Var}(X) = \text{variance of } X, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

Expected value

The symbol $\mathbb{E}[X]$ means the long-run average, or expected value, of X under the model. It does not mean that every individual observation equals that value. A zero-mean error can still be positive on one record and negative on another.

A normal distribution

The notation

$$U \sim \mathcal{N}(0, \sigma^2)$$

means that the random error U is modeled by a bell-shaped **normal distribution** centered at zero. The first number, 0, is its mean. The second number, σ^2 , is its variance. Therefore σ is its standard deviation.

The symbol “ $\sim$ ” here means “is modeled as following the distribution.” It does not mean “approximately equal to.”

Conditional notation

The vertical bar in $p(w | x)$ is read as “given.” Thus $p(w | x)$ describes how plausible a recorded value w is *given that* the clean value is x .

Likewise, $\mathbb{E}[W | X = x]$ means the expected recorded value when we hold the clean value fixed at x .

Indicator: an on/off switch written mathematically

Later we need a term that is 1 for target-group rows and 0 for reference-group rows. We write

$$\mathbf{1}[g_i = \text{tgt}] = \begin{cases} 1, & \text{if row } i \text{ belongs to the target group,} \\ 0, & \text{otherwise.} \end{cases}$$

This is called an **indicator**. It is simply a mathematical on/off switch.

Absolute value

The notation $|a|$ means the distance of a from zero. Thus $|3| = 3$ and $|-3| = 3$. It is useful when we care about the size of a change but not its direction.

A vector and its Euclidean length

A **vector** is simply an ordered list of numbers, for example $(0.07, -0.20)$. Its ordinary geometric length is

$$\|(0.07, -0.20)\|_2 = \sqrt{0.07^2 + (-0.20)^2}.$$

This is the same Pythagorean distance used for a right triangle, extended to any number of coordinates. The subscript 2 indicates this ordinary Euclidean length.

Exponentials and the number e

The function e^{-x} starts at 1 when $x = 0$ and decreases toward 0 as x grows. For example,

$$e^0 = 1, \quad e^{-1} \approx 0.368, \quad e^{-2} \approx 0.135.$$

This makes e^{-x} a convenient way to turn a nonnegative penalty into a weight between 0 and 1.

The symbol $\propto$

The symbol $\propto$ means “proportional to.” If we write

$$p(z) \propto f(z),$$

it means that $p(z) = Cf(z)$ for some positive constant C chosen so that all probabilities add or integrate to 1. We often use this notation when the shape of a probability distribution is known but its normalizing constant is inconvenient to calculate directly.

Data notation used throughout

Rows are indexed by $i = 1, \dots, N$. Features are indexed by $j = 1, \dots, p$. A feature might be carbon intensity, durability, repairability, or another numerical attribute.

Symbol	Meaning in ordinary language
y_{ij}	the value actually recorded in row i for feature j
x_{ij}	the cleaner underlying value that the method is trying to infer
g_i	the group membership of row i : reference or target
$\Delta_j(z)$	the clean target-group mean minus the clean reference-group mean for feature j under one proposed hidden state z
$g_j^\star$	the external anchor: the expected clean target-minus-reference gap for feature j
$h_j(z)$	a scale, usually a standard deviation, used so features with different units can be compared fairly
G	the set of features for which a gap anchor is actively enforced

A **hidden state** z means one complete proposed explanation of the unobserved clean table and associated bias quantities. The statistical word often used for “hidden” is *latent*. We will mostly say “hidden” or “unobserved” instead.

The document always defines a group gap as

$$\text{target-group mean} - \text{reference-group mean}.$$

The order matters. Reversing it would reverse the sign of every anchor.

Equation (1): recorded value = clean value + ordinary error

Start with the simplest possible idea

Suppose there is an unknown clean value X and the database records a value W . The recording error is simply the difference

$$U = W - X.$$

Rearranging this definition gives

$$W = X + U.$$

That is the core of Equation (1): the recorded value equals the clean value plus an error.

Add a model for the error

The paper assumes the ordinary row-level error is centered at zero and is approximately bell-shaped. This is written as

$$W = X + U, \quad U \sim \mathcal{N}(0, \sigma^2). \quad (1)$$

In plain English. W means the number that is actually recorded. X means the cleaner underlying value we wish we could observe directly. U is simply the difference between those two quantities.

New words used here. **Random error** means a disturbance that can vary from one measurement to another. **Normal distribution** means the familiar bell-shaped probability model. **Zero mean** means positive and negative errors balance to zero on average under the model; it does not mean every error is zero.

What Equation (1) immediately implies

If the clean value is fixed at $X = x$, then $W = x + U$. Adding a fixed number shifts the center of a distribution but does not change its spread. Therefore

$$W \mid X = x \sim \mathcal{N}(x, \sigma^2).$$

The expected recorded value is

$$\mathbb{E}[W \mid X] = X,$$

and its conditional variance is

$$\text{Var}(W \mid X) = \sigma^2.$$

So, under this model, ordinary random error does not systematically push the expected recorded value up or down.

Where the probability-density formula comes from

The normal distribution with center x and standard deviation σ has the density

$$p(w \mid x, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(w-x)^2}{2\sigma^2}\right]. \quad (1a)$$

A **density** is a curve describing where continuous values are concentrated. The factor $(w-x)^2$ is small when the recorded value w is close to the proposed clean value x , so such an x receives greater support than a far-away value, all else equal.

Numerical example

Let the clean value be $X = 70$, let the error standard deviation be $\sigma = 3$, and suppose one realized error is $U = +2.4$. Then

$$W = 70 + 2.4 = 72.4.$$

The error is

$$\frac{W - X}{\sigma} = \frac{2.4}{3} = 0.8$$

standard deviations above the clean value.

If we average many independent zero-mean errors, their random fluctuations tend to cancel. In contrast, a common shift shared by an entire group does *not* disappear by averaging. That motivates Equation (2).

Mathematical check. The algebra $U = W - X \iff W = X + U$ is exact. Under $\mathbb{E}[U] = 0$, $\mathbb{E}[W \mid X] = X$. Under $\text{Var}(U) = \sigma^2$, $\text{Var}(W \mid X) = \sigma^2$. The displayed normal density is the standard normal-density formula shifted to mean x .

What this does not prove. Additivity, zero-mean error, and normality are assumptions about the measurement process. The equation does not prove that a real data source satisfies those assumptions.

Equation (2): adding a shared bias for one group

Why Equation (1) is not enough

Equation (1) describes ordinary error that changes from one record to another. Now consider a different situation: every record in one group is shifted in the same direction. Averaging does not remove that shared shift.

We therefore split total error into two parts:

1. a systematic group shift, shared by target-group records for a feature; and
2. an ordinary row-specific random error.

Build the systematic shift one piece at a time

For feature j , let $b_j \geq 0$ be the size of the systematic shift. We keep b_j nonnegative so it describes magnitude only.

Let

$$d_j \in \{-1, +1\}$$

carry the direction. If $d_j = +1$, the systematic shift is upward; if $d_j = -1$, it is downward.

Finally, use the indicator $\mathbf{1}[g_i = \text{tgt}]$ as an on/off switch. It is 1 for target rows and 0 for reference rows. The systematic part is therefore

$$S_{ij} = d_j b_j \mathbf{1}[g_i = \text{tgt}].$$

For a reference row this becomes 0. For a target row it becomes $d_j b_j$.

Add an ordinary error ε_{ij} and we obtain

$$y_{ij} = x_{ij} + d_j b_j \mathbf{1}[g_i = \text{tgt}] + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_j^2). \quad (2)$$

In plain English. For each table cell, Equation (2) says: start from the hidden clean value; if the row belongs to the target group, add the shared signed bias; then add ordinary record-level noise. Reference rows do not receive the shared target-group shift.

New words used here. **Systematic bias** here means a shift that repeatedly acts in the same direction for a group. It is different from random error, which can move individual observations up or down.

Deriving Equation (2) from Equation (1)

Start with the generic additive-error form

$$y_{ij} = x_{ij} + U_{ij}.$$

Write total error as

$$U_{ij} = S_{ij} + \varepsilon_{ij}, \quad S_{ij} = d_j b_j \mathbf{1}[g_i = \text{tgt}]. \quad (2a)$$

Substituting the second line into the first gives Equation (2) exactly.

What happens to the average of each group?

For a reference row, the indicator is 0, so

$$\mathbb{E}[y_{ij} \mid x_{ij}, g_i = \text{ref}] = x_{ij}.$$

For a target row, the indicator is 1, so

$$\mathbb{E}[y_{ij} \mid x_{ij}, g_i = \text{tgt}] = x_{ij} + d_j b_j.$$

Averaging within each group gives

$$\mathbb{E}[\bar{y}_{\text{tgt},j} \mid X] = \bar{x}_{\text{tgt},j} + d_j b_j, \quad \mathbb{E}[\bar{y}_{\text{ref},j} \mid X] = \bar{x}_{\text{ref},j}.$$

Subtract the reference-group mean from the target-group mean:

$$\mathbb{E}[\bar{y}_{\text{tgt},j} - \bar{y}_{\text{ref},j} \mid X] = (\bar{x}_{\text{tgt},j} - \bar{x}_{\text{ref},j}) + d_j b_j.$$

Therefore

$$\text{expected recorded gap} = \text{clean gap} + \text{systematic shift}.$$

Numerical example

Suppose the clean reference values are

$$68, 72, 70,$$

with mean 70, and the clean target values are

$$74, 76, 75,$$

with mean 75. The clean target-minus-reference gap is therefore $75 - 70 = 5$.

Let $d = +1$ and $b = 5$. For the three target records, use ordinary errors -1 , $+0.5$, and 0 . Their recorded values become

$$74 + 5 - 1 = 78, \quad 76 + 5 + 0.5 = 81.5, \quad 75 + 5 + 0 = 80.$$

Their recorded mean is

$$\frac{78 + 81.5 + 80}{3} = 79.833 \dots$$

and the recorded group gap is

$$79.833 \dots - 70 = 9.833 \dots$$

The expected gap without the small random errors would be $5 + 5 = 10$.

Why outside information is unavoidable

If all we observe is a recorded gap of about 10, we cannot tell whether the clean gap is 10 and the bias is 0, the clean gap is 5 and the bias is 5, or some other combination. This is a mathematical non-identifiability problem, not a limitation that can be solved merely by collecting more records with the same shared bias.

Mathematical check. The target/reference expectation equations follow directly from the indicator taking values 1 and 0. Subtracting the two expected group means gives the clean gap plus $d_j b_j$. The numerical example reproduces a recorded target mean of $79.833 \dots$ and a recorded gap of $9.833 \dots$.

What this does not prove. Equation (2) assumes the shared target shift is additive, that the declared direction d_j is correct, that the reference group does not have an analogous shared shift, and that the residual errors are adequately represented by the stated noise model. Later stress tests deliberately violate some of these assumptions.

Equation (3): reweighting possible hidden explanations

The problem Equation (3) addresses

After Equation (2), many different hidden clean tables can still be compatible with the observed records. Instead of pretending that one hidden table is certainly correct, the method keeps a collection of plausible possibilities and gives different possibilities different weights.

This is easiest to understand with a small finite example. Imagine only three possible hidden explanations: A, B, and C. Before applying the paper’s gap and bias penalties, suppose their weights are

$$0.50, \quad 0.30, \quad 0.20.$$

These sum to 1, so they can be interpreted as a probability distribution over the three possibilities.

New words used here. A **probability distribution over hidden states** is simply a set of weights over possible hidden explanations. In continuous problems there can be infinitely many possibilities, so the weights are described by a density rather than a short list.

Start from a base distribution

Let z denote one complete hidden state: a proposed clean table together with the bias quantities needed by the model. Let

$$p_0(z \mid y, g^\star)$$

be the **base distribution**: the model’s starting weight for state z after using the observed data y , the model assumptions, and the anchor information already included in the implementation.

The subscript 0 means “base” or “starting” distribution for the additional penalty step; it does not mean a probability of zero.

Turn disagreements into penalties

The method defines two nonnegative penalty scores:

- $C_{\text{gap}}(z)$: how far the clean group gaps in state z are from the anchor values, after accounting for scale;
- $C_{\text{bias}}(z)$: how much state z leaves disparity in a declared adverse direction.

Equations (4) and (5) explain these penalties from first principles.

The constants $\lambda_g \geq 0$ and $\lambda_b \geq 0$ control how strongly the two penalties matter. Combine them into one total loss:

$$L(z) = \lambda_g C_{\text{gap}}(z) + \lambda_b C_{\text{bias}}(z).$$

A larger $L(z)$ means the state violates the desired conditions more strongly.

Convert the penalty into a downweighting factor

Because $L(z) \geq 0$, the quantity

$$e^{-L(z)}$$

is between 0 and 1. If $L(z) = 0$, the factor is 1 and the base weight is unchanged. If $L(z) = 2$, the factor is $e^{-2} \approx 0.135$, so the state is strongly downweighted.

Multiply the base weight by this factor:

$$\text{new unnormalized weight} = p_0(z \mid y, g^\star) e^{-L(z)}.$$

After dividing by the total weight over all possible states so that the probabilities again sum or integrate to 1, we obtain the corrected distribution. The compact notation is

$$\pi_\lambda(z \mid y, g^\star) \propto p_0(z \mid y, g^\star) \exp\{-\lambda_g C_{\text{gap}}(z) - \lambda_b C_{\text{bias}}(z)\}. \quad (3)$$

In plain English. Equation (3) does not create a new candidate clean table. It changes the relative weights of candidate hidden states. A state that already looked plausible and agrees well with the anchor-related conditions receives more relative weight than a state with large penalties.

New words used here. **Generalized Bayes** is a way of updating uncertainty using a loss function rather than requiring every piece of information to be written as an ordinary probability likelihood (a formula describing how plausible the observed data would be under a proposed state). **Exponential tilt** means multiplying the original distribution by an exponential penalty factor and then renormalizing.

A concrete three-state calculation

Use base probabilities

$$p_0 = (0.50, 0.30, 0.20),$$

loss weights $\lambda_g = 1.2$ and $\lambda_b = 0.8$, and penalty pairs

$$(C_{\text{gap}}, C_{\text{bias}}) = (0, 0.2), (1, 0.5), (2, 1).$$

The three total losses are

$$0.16, \quad 1.60, \quad 3.20.$$

The exponential factors are approximately

$$e^{-0.16} = 0.8521, \quad e^{-1.60} = 0.2019, \quad e^{-3.20} = 0.04076.$$

Multiplying by the base probabilities gives unnormalized weights

$$0.4261, \quad 0.06057, \quad 0.00815.$$

Their sum is about 0.4948. Divide each by that sum:

$$\pi_\lambda \approx (0.8611, 0.1224, 0.0165).$$

The penalties do not merely subtract a fixed amount from every probability. They change the *relative* support for the competing hidden explanations.

Why this exponential form has a principled derivation

The preceding explanation is enough to use Equation (3), but the exponential form can also be derived by asking for a new distribution q that balances two goals:

1. prefer states with small expected loss; and
2. avoid moving unnecessarily far from the base distribution p_0 .

The mathematical measure of the second goal is the Kullback–Leibler divergence,

$$\text{KL}(q \| p_0).$$

It is nonnegative and is zero when q and p_0 are the same distribution. The optimization problem is

$$q^* = \arg \min_q \{ \mathbb{E}_q [L(z)] + \text{KL}(q \| p_0) \}. \quad (3a)$$

Here $\mathbb{E}_q [L(z)]$ means the average loss when states are weighted according to q .

New words used here. **KL divergence** is a measure of how much one probability distribution differs from another. $\arg \min$ means “the choice of q that makes the following expression as small as possible.”

For readers who want the algebra: enforcing the requirement that q integrates to 1 and differentiating the objective with respect to the whole function q gives

$$\log q(z) = \log p_0(z) - L(z) - \text{constant}.$$

Exponentiating both sides gives

$$q(z) \propto p_0(z) e^{-L(z)},$$

which is exactly Equation (3).

How samples are obtained when direct calculation is difficult

The corrected distribution may be too complicated to normalize and sample from directly. The implementation therefore uses a Markov chain Monte Carlo method, specifically a Metropolis–Hastings-style procedure. The early part of such a computer sequence is used to let the calculation settle; the later states that are kept for summaries are called *retained states*.

The basic idea is simple: start from one hidden state z , propose another state z' , and decide whether to move there. For a proposal distribution $q(z' | z)$, the textbook acceptance probability is

$$\alpha(z \rightarrow z') = \min \left\{ 1, \frac{\pi(z')q(z | z')}{\pi(z)q(z' | z)} \right\}. \quad (3b)$$

If the proposal mechanism is symmetric, meaning proposing z' from z is as likely as proposing z from z' , the two proposal terms cancel.

New words used here. **MCMC** stands for Markov chain Monte Carlo. It generates a sequence of dependent samples whose long-run distribution is intended to match a target distribution. **Metropolis–Hastings** is one standard MCMC rule for accepting or rejecting proposed moves.

Mathematical check. The finite-state reweighting numbers were recalculated. The optimization problem $\mathbb{E}_q[L] + \text{KL}(q \| p_0)$ has solution $q \propto p_0 e^{-L}$ provided the required integrals exist and the distribution can be normalized. This matches the generalized-Bayes loss-update form in Bissiri, Holmes, and Walker. The Metropolis–Hastings acceptance formula shown in (3b) is the standard general form.

What this does not prove. Equation (3) defines the distribution the computer is intended to represent. The actual sequence of computer-generated states is only as accurate as its proposal rules and as the extent to which the sequence has settled into that intended distribution. The released sampler includes bounded/clipped moves and is therefore described conservatively as approximate rather than as a mathematically proven exact sampler from the intended final distribution.

Equation (4): measuring how far clean gaps are from their anchors

First define the gap for one feature

For feature j , let

$$\Delta_j(z) = \text{clean target mean} - \text{clean reference mean}$$

under hidden state z . Let $g_j^\star$ be the corresponding anchor. Their raw difference is

$$r_j = \Delta_j(z) - g_j^\star.$$

If $r_j = 0$, the state exactly matches the anchor for that feature. A positive or negative r_j means it differs in one direction or the other.

Why raw differences cannot simply be mixed across features

Suppose carbon intensity is measured in numbers around tens while repairability is a score around 0–10. A raw difference of 5 in carbon units and a raw difference of 1 in repairability points cannot be compared fairly by magnitude alone. They use different units and natural scales.

The method therefore divides each raw mismatch by a scale $h_j(z)$ that has the same unit as feature j :

$$u_j = \frac{\Delta_j(z) - g_j^\star}{h_j(z)}.$$

After division, u_j has no physical unit. A value $u_j = 0.2$ means the gap differs from its anchor by one fifth of the chosen scale, regardless of the original unit.

New words used here. **Standardization** means dividing a quantity by an appropriate scale so the result becomes unit-free and can be compared more fairly with other standardized quantities. In the implementation, h_j is based on the completed column's standard deviation.

Combine several standardized mismatches

For all anchored features $j \in G$, collect the standardized mismatches into a vector and take its ordinary Euclidean length:

$$C_{\text{gap}}(z) = \left\| \left(\frac{\Delta_j(z) - g_j^*}{h_j(z)} \right)_{j \in G} \right\|_2. \quad (4)$$

Using the definition of Euclidean length, this is equivalent to

$$C_{\text{gap}}^2(z) = \sum_{j \in G} \left(\frac{\Delta_j(z) - g_j^*}{h_j(z)} \right)^2. \quad (4a)$$

In plain English. Equation (4) asks: after putting every anchored feature on a comparable unit-free scale, how far is this hidden state's whole set of group gaps from the anchor point? It is just a Pythagorean distance in several dimensions.

Why both positive and negative mismatches count

The residual is squared inside the Euclidean length. Therefore $+0.2$ and -0.2 contribute equally:

$$(+0.2)^2 = (-0.2)^2 = 0.04.$$

Equation (4) is therefore **two-sided**: it penalizes disagreement in either direction.

Numerical example

Consider two anchored features. Let

$$\Delta = (5.7, -0.6), \quad g^* = (5.0, -0.3), \quad h = (10, 1.5).$$

The raw residuals are

$$5.7 - 5.0 = 0.7, \quad -0.6 - (-0.3) = -0.3.$$

Standardize them:

$$0.7/10 = 0.07, \quad -0.3/1.5 = -0.20.$$

Then

$$C_{\text{gap}} = \sqrt{0.07^2 + (-0.20)^2} = \sqrt{0.0049 + 0.0400} \approx 0.2119. \quad (4b)$$

Although 0.3 is a smaller raw number than 0.7, the second mismatch contributes more because it is large relative to its own feature scale.

What happens if the scale is zero?

Division by $h_j = 0$ is undefined. A column with essentially no spread therefore needs a safeguard. The implementation replaces a non-finite or extremely small scale with a safe fallback value. This is an engineering guard against division by zero; it does not make a zero-variance feature statistically informative by itself.

Mathematical check. For every $h_j > 0$, each standardized residual is unit-free, its square is nonnegative, and therefore $C_{\text{gap}} \geq 0$. The score is exactly zero if and only if every constrained residual is zero. The numerical result is $\sqrt{0.0449} = 0.211896 \dots$

What this does not prove. The Euclidean norm ignores the possibility that two features systematically move together. A distance that explicitly accounts for such co-movement would require a stable estimate of how the features vary together; Equation (4) does not use that extra information.

Equation (5): penalizing only movement in a declared adverse direction

Why Equation (5) is different from Equation (4)

Equation (4) treats a mismatch in either direction as a mismatch. Equation (5) asks a different question: does the clean gap lie beyond the anchor specifically in a direction that the domain definition has declared adverse?

For example, larger carbon emissions may be considered adverse, while a lower protective or repairability score may be adverse. The direction must be supplied from feature semantics; mathematics does not decide what is socially or physically “good” or “bad.”

Step 1: compute the residual

As before,

$$r_j = \Delta_j(z) - g_j^\star.$$

Step 2: orient the residual

Choose

$$d_j \in \{-1, +1\}.$$

The convention is chosen so that

$$a_j = d_j r_j > 0$$

means “the residual points in the declared adverse direction.”

If $d_j = +1$, then larger-than-anchor gaps are adverse. If $d_j = -1$, then smaller-than-anchor gaps are adverse.

Step 3: keep only adverse residuals

Use

$$\max\{0, a_j\}.$$

This returns a_j if it is positive and 0 otherwise. For example,

$$\max\{0, 0.4\} = 0.4, \quad \max\{0, -0.4\} = 0.$$

This operation is sometimes called a *hinge* or *positive-part* function. Nothing more complicated is meant: negative values are clipped to zero.

Step 4: standardize and average

Divide by $h_j(z)$ to remove units, then average over all p modeled features. The result is

$$C_{\text{bias}}(z) = \frac{1}{p} \sum_{j=1}^p \frac{\max\{0, d_j(\Delta_j(z) - g_j^*)\}}{h_j(z)}. \quad (5)$$

In plain English. For each feature, Equation (5) asks only whether the gap exceeds its anchor in the feature's declared adverse direction. If not, that feature contributes zero. If yes, the size of the adverse excess is standardized and added to the average.

Reading the direction sign carefully

When $d_j = +1$,

$$\max\{0, d_j(\Delta_j - g_j^*)\} = \max\{0, \Delta_j - g_j^*\},$$

so a penalty appears only when $\Delta_j > g_j^*$.

When $d_j = -1$,

$$\max\{0, d_j(\Delta_j - g_j^*)\} = \max\{0, g_j^* - \Delta_j\},$$

so a penalty appears only when $\Delta_j < g_j^*$.

Numerical example

Let

$$p = 3, \quad d = (+1, -1, +1), \quad \Delta = (6.0, -0.1, 0.3), \\ g^* = (5.0, -0.3, 0), \quad h = (10, 1, 0.5).$$

For feature 1,

$$r_1 = 1.0, \quad d_1 r_1 = 1.0, \quad \frac{\max(0, 1.0)}{10} = 0.10.$$

For feature 2,

$$r_2 = 0.2, \quad d_2 r_2 = -0.2, \quad \frac{\max(0, -0.2)}{1} = 0.$$

For feature 3,

$$r_3 = 0.3, \quad d_3 r_3 = 0.3, \quad \frac{\max(0, 0.3)}{0.5} = 0.60.$$

Average the three contributions:

$$C_{\text{bias}} = \frac{0.10 + 0 + 0.60}{3} = 0.2333 \dots \quad (5a)$$

Mathematical check. Every numerator in Equation (5) is nonnegative because of $\max(0, \cdot)$. With positive scales h_j , every summand is therefore nonnegative, so $C_{\text{bias}} \geq 0$. The terms are unit-free after standardization. The worked example evaluates to $0.7/3 = 0.233333 \dots$

What this does not prove. The direction signs d_j come from the real-world meaning assigned to each feature and from the system's domain rules or configuration. Equation (5) cannot verify that those choices are substantively correct. Also, averaging can make one severe violation look smaller when many other features contribute zero; that is a design consequence of using a mean rather than a maximum or a weighted sum.

Equation (6): building an uncertainty interval

What an interval is trying to communicate

A single estimate such as 5.30 can suggest more precision than the analysis actually has. An uncertainty interval gives a center together with a range of values that reflects modeled uncertainty.

Let $\hat{x}$ be the average of the retained estimates of the clean quantity of interest (the later MCMC states kept after the initial settling period). The hat over x means “estimate.” Let $\hat{v}$ be the variance of those retained values.

The method also includes an external residual-bias cushion σ_b . This is intended to represent remaining bias uncertainty not already captured by the retained-state variation.

New words used here. **Residual bias** means bias that may remain after the main correction process. In this method, σ_b is supplied externally; it is not learned automatically from the chain.

Why the variances are added

Let Q represent uncertainty already present in the retained query values, with

$$\text{Var}(Q) = \hat{v}.$$

Let B represent an additional zero-mean residual-bias uncertainty with

$$\text{Var}(B) = \sigma_b^2.$$

For any two random quantities,

$$\text{Var}(Q + B) = \text{Var}(Q) + \text{Var}(B) + 2 \text{Cov}(Q, B),$$

where covariance measures whether the two uncertainties tend to move together.

If the method assumes Q and B are independent, then their covariance is zero. Therefore

$$\text{Var}(Q + B) = \hat{v} + \sigma_b^2.$$

Taking the square root returns to the original unit:

$$\text{SD}_{\text{total}} = \sqrt{\hat{v} + \sigma_b^2}.$$

New words used here. **Independent** means knowing one of the two random quantities does not provide information about the other under the model. **Covariance** measures whether two quantities tend to vary together. Independence implies zero covariance, although zero covariance does not always imply full independence.

Where 1.96 comes from

For an exactly normal distribution, about 95% of probability lies between roughly -1.96 and $+1.96$ standard deviations from the mean. Thus a normal-style central interval has half-width

$$1.96 \times \text{SD}_{\text{total}}.$$

This leads to

$$I = \hat{x} \pm 1.96 \sqrt{\hat{v} + \sigma_b^2}. \quad (6)$$

Equivalently, if L and U denote the lower and upper endpoints,

$$\begin{aligned} \text{SD}_{\text{total}} &= \sqrt{\hat{v} + \sigma_b^2}, \\ L &= \hat{x} - z_{0.975} \text{SD}_{\text{total}}, \\ U &= \hat{x} + z_{0.975} \text{SD}_{\text{total}}, \quad z_{0.975} \approx 1.96. \end{aligned} \tag{6a}$$

Here $z_{0.975}$ means the point below which 97.5% of a standard normal distribution lies.

In plain English. Equation (6) says: combine the two variance components, convert the result back to a standard deviation, multiply by 1.96, and extend that distance below and above the estimated center.

Numerical example

Take

$$\hat{x} = 5.30, \quad \hat{v} = 0.16, \quad \sigma_b = 0.75.$$

First square the residual-bias standard deviation:

$$\sigma_b^2 = 0.75^2 = 0.5625.$$

Add the variances:

$$0.16 + 0.5625 = 0.7225.$$

Take the square root:

$$\sqrt{0.7225} = 0.85.$$

Multiply by 1.96:

$$1.96 \times 0.85 = 1.666.$$

Therefore

$$L = 5.30 - 1.666 = 3.634, \quad U = 5.30 + 1.666 = 6.966.$$

So

$$I = [3.634, 6.966], \quad w = U - L = 3.332. \tag{6b}$$

The symbol w is the **full width** of the interval, not the half-width. That convention matters in Equation (9).

Mathematical check. The general variance identity includes a covariance term. Under the stated independence assumption, that term is zero and the variance addition in Equation (6) is correct. The numerical square root is exactly 0.85, the half-width is 1.666, and the full width is 3.332. The standard-normal 0.975 quantile is approximately 1.96.

What this does not prove. This is a nominal normal-style 95% uncertainty interval. It is not a theorem guaranteeing that intervals built this way would contain the true value exactly 95% of the time over repeated studies, and it is not an exact interval derived from a fully specified probability-update model. The residual-bias term is externally chosen, the retained-state distribution is approximate, and the combined uncertainty need not be perfectly normal.

Equation (7): asking how much the answer depends on the anchor

The idea before the formula

An anchor helps the model separate a possible clean group difference from systematic reporting bias. Because the anchor matters, we should also ask a simple robustness question:

If we deliberately move the anchor by a known amount and rerun the estimator, how much does the estimated answer move?

Let $\hat{x}(g^*)$ be the estimated answer using the original anchor g^* . Change the anchor to $g^* + \delta$, where δ is a predeclared perturbation, and recompute the estimate as $\hat{x}(g^* + \delta)$.

The absolute change is

$$s = |\hat{x}(g^*) - \hat{x}(g^* + \delta)|. \quad (7)$$

In plain English. Equation (7) is a controlled “what-if” test. Change the anchor, rerun the method, and measure how much the answer changes. A larger s means the answer is more dependent on that particular anchor perturbation.

New words used here. **Sensitivity analysis** means deliberately changing an input or assumption and observing how much the result changes. A **perturbation** is simply the deliberate change.

Why absolute value is used

Suppose the original estimate is 5.3. If the changed anchor makes the estimate 6.1, the movement is +0.8. If another perturbation made it 4.5, the movement would be −0.8. In both cases the size of the movement is 0.8, so absolute value is appropriate when the diagnostic is intended to measure magnitude.

Relation to a slope or derivative

For a scalar anchor, the ratio

$$\frac{\hat{x}(g^* + \delta) - \hat{x}(g^*)}{\delta}$$

is a finite-difference approximation to the local slope of the estimated answer with respect to the anchor when δ is small.

Equation (7) intentionally does *not* divide by δ . The reported score asks for the actual answer change caused by a fixed, predeclared perturbation. If two experiments use different values of δ , their raw s values are not directly comparable unless the perturbation is normalized.

New words used here. A **derivative** is the mathematical idea of a local slope: how fast an output changes when an input changes by a very small amount. A **finite difference** uses an ordinary nonzero change, such as +2, to approximate or summarize that behavior.

Numerical example

Let

$$g^* = 5.0, \quad \delta = 2.0,$$

and suppose

$$\hat{x}(g^*) = 5.30, \quad \hat{x}(g^* + \delta) = 6.10.$$

Then

$$s = |5.30 - 6.10| = 0.80. \quad (7a)$$

If we also want a change-per-unit-anchor quantity, it is

$$0.80/2.0 = 0.40.$$

Mathematical check. Equation (7) is exactly the absolute finite change between two estimator runs. Its unit is the same as the answer because it subtracts two answers in the same unit. The worked example gives $s = 0.80$, and dividing by $\delta = 2$ gives a normalized finite-difference magnitude of 0.40.

What this does not prove. A small s does not prove that the anchor is correct. It only says the estimate does not change much under the particular perturbation tested. A large s does not prove the anchor is wrong either; it says the answer depends strongly on that input in the tested neighborhood.

Equation (8): combining an audit estimate with a prior anchor

Why combine two sources?

Suppose a small manual audit gives one estimate of the clean group gap, while an existing prior anchor gives another. A small audit is directly relevant but may be noisy because it contains few verified records. The prior anchor may be more stable but less specific to the current sample.

Equation (8) forms a weighted average of these two sources.

Let

- $\bar{g}_{\text{audit},j}$ = the gap estimated from verified audit records for feature j ;
- $g_{\text{prior},j}$ = the prior anchor for feature j ;
- n_j = the effective audit count used for feature j ;
- k = a fixed amount of weight given to the prior anchor.

In the reported setup, $k = 12$.

Build the weighted average

Give the audit value weight n_j and the prior value weight k . Multiply each value by its weight, add the results, and divide by the total weight:

$$g_j^\star = \frac{n_j \bar{g}_{\text{audit},j} + k g_{\text{prior},j}}{n_j + k}, \quad k = 12. \quad (8)$$

In plain English. Equation (8) is the same basic structure as an ordinary weighted average. More verified audit records move the result toward the audit gap. With very little audit information, the prior anchor receives more relative influence.

New words used here. **Shrinkage** means pulling a noisy estimate toward a more stable reference value. A **pseudocount** is a convenient way of expressing prior weight as if it contributed a fixed number of observations. Here $k = 12$ is an engineering weight; it does not mean that 12 additional real records were observed.

Make the weights visible

Define

$$w_{\text{audit}} = \frac{n_j}{n_j + k}, \quad w_{\text{prior}} = \frac{k}{n_j + k}.$$

Then

$$w_{\text{audit}} + w_{\text{prior}} = \frac{n_j + k}{n_j + k} = 1.$$

Equation (8) can therefore be written as

$$g_j^\star = w_{\text{audit}} \bar{g}_{\text{audit},j} + w_{\text{prior}} g_{\text{prior},j}. \quad (8a)$$

When $n_j \geq 0$, $k \geq 0$, and $n_j + k > 0$, both weights are nonnegative and sum to 1. Therefore the combined anchor must lie between the two input values.

Another derivation: minimize disagreement with both sources

Equation (8) can also be derived by choosing a value g that keeps weighted squared disagreement with both sources as small as possible:

$$J(g) = n_j (g - \bar{g}_{\text{audit},j})^2 + k (g - g_{\text{prior},j})^2. \quad (8b)$$

The function $J(g)$ is smallest where its slope is zero. Differentiate:

$$J'(g) = 2n_j(g - \bar{g}_{\text{audit},j}) + 2k(g - g_{\text{prior},j}).$$

Set this equal to zero:

$$2n_j(g - \bar{g}_{\text{audit},j}) + 2k(g - g_{\text{prior},j}) = 0.$$

Divide by 2 and collect the terms containing g :

$$(n_j + k)g = n_j\bar{g}_{\text{audit},j} + kg_{\text{prior},j}.$$

Finally divide by $n_j + k$ to recover Equation (8).

Why this resembles a common probability-updating formula, but should not be oversold

In a common probability model using normal distributions, the updated mean can have the same weighted-average form, with more precise sources (those with smaller variance) receiving more weight. Here, however, k is not estimated from such a fully specified variance model. Calling it a pseudocount is more precise: it is a fixed design choice that controls how strongly the prior anchor pulls the audit estimate.

Numerical example

Let

$$\bar{g}_{\text{audit}} = 5.6, \quad g_{\text{prior}} = 5.0, \quad n = 20, \quad k = 12.$$

Then

$$g^* = \frac{20(5.6) + 12(5.0)}{20 + 12} = \frac{112 + 60}{32} = \frac{172}{32} = 5.375. \quad (8c)$$

The audit weight is $20/32 = 0.625$ and the prior weight is $12/32 = 0.375$.

As $n \rightarrow 0$, the audit receives almost no weight and $g^* \rightarrow g_{\text{prior}}$. As n becomes very large, the audit dominates and $g^* \rightarrow \bar{g}_{\text{audit}}$.

Mathematical check. The two weights are nonnegative and sum exactly to 1 whenever $n_j, k \geq 0$ and $n_j + k > 0$. Differentiating the quadratic objective and solving $J'(g) = 0$ reproduces Equation (8). Because $J''(g) = 2(n_j + k) > 0$, the point where the slope is zero is the unique minimum, not a maximum. The numerical result is exactly 5.375.

What this does not prove. A weighted average can stabilize a noisy audit, but it cannot prove that either the audit sample or the prior anchor represents the correct population, time period, feature definition, or measurement process.

Equation (9): combining evidence, interval width, and anchor sensitivity into an answer/abstain score

What decision is being made?

The system eventually has to choose between two actions:

- answer the question; or
- abstain, meaning deliberately withhold an answer because the available support is judged insufficient.

Equation (9) builds a single score for that decision from three ingredients.

Let

- c_{ev} = an evidence-availability score;

- w = the full uncertainty-interval width from Equation (6);
- s = the anchor-sensitivity magnitude from Equation (7).

A wider interval means more uncertainty. A larger sensitivity means the answer changes more when the anchor is perturbed. Both should reduce the final selection score.

Why divide by w_0 and s_0 ?

The quantities w and s have units. We cannot add “3 carbon units” directly to “0.8 repairability units” and claim the sum has a universal meaning. Positive scale constants w_0 and s_0 convert each quantity into a unit-free ratio:

$$\frac{w}{w_0}, \quad \frac{s}{s_0}.$$

Build a nonnegative danger score

Average the two unit-free ratios:

$$D = \frac{1}{2} \left(\frac{w}{w_0} + \frac{s}{s_0} \right).$$

Because $w, s \geq 0$ and $w_0, s_0 > 0$, we have $D \geq 0$.

Convert danger into a decreasing trust factor

Use

$$\text{trust} = e^{-D}.$$

When $D = 0$, trust is 1. As D grows, e^{-D} decreases smoothly toward 0.

Multiply this trust factor by the evidence score:

$$c = c_{\text{ev}} \times \text{trust}.$$

Combining the steps gives the paper’s equation

$$c = c_{\text{ev}} \exp \left[-\frac{1}{2} \left(\frac{w}{w_0} + \frac{s}{s_0} \right) \right]. \quad (9)$$

Equivalently,

$$D = \frac{1}{2} \left(\frac{w}{w_0} + \frac{s}{s_0} \right), \quad \text{trust} = e^{-D}, \quad c = c_{\text{ev}} \text{ trust}. \quad (9a)$$

In plain English. Equation (9) starts with the available-evidence score and reduces it when the uncertainty interval is wide or the answer is strongly anchor-sensitive. It is a decision score, not a probability that the answer is correct.

New words used here. **Abstention** means the system intentionally does not provide a substantive answer. **Monotone decreasing** means that increasing an input such as interval width cannot make the score go up when everything else is held fixed.

Boundary checks

If $w = 0$ and $s = 0$, then $D = 0$ and

$$c = c_{\text{ev}}.$$

Thus the trust adjustment never increases the evidence score.

If $c_{ev} \in [0, 1]$, then because $0 < e^{-D} \leq 1$,

$$0 \leq c \leq c_{ev} \leq 1.$$

The implementation additionally clips the final score to $[0, 1]$.

If either w or s becomes very large while the scales remain fixed, then D becomes large and c approaches 0.

Checking the direction mathematically

Treat all other quantities as fixed. Differentiating gives

$$\frac{\partial c}{\partial w} = -\frac{c}{2w_0} \leq 0, \quad \frac{\partial c}{\partial s} = -\frac{c}{2s_0} \leq 0.$$

So increasing interval width or anchor sensitivity cannot increase c under the formula.

Numerical example

Use

$$c_{ev} = 0.85, \quad w = 3.332, \quad w_0 = 4.0, \quad s = 0.80, \quad s_0 = 1.20.$$

Then

$$D = \frac{1}{2} \left(\frac{3.332}{4} + \frac{0.80}{1.20} \right) \approx 0.7498.$$

Therefore

$$\text{trust} = e^{-0.7498} \approx 0.4724,$$

and

$$c = 0.85(0.4724) \approx 0.4016. \tag{9b}$$

With threshold $\tau = 0.35$, the rule answers because $0.4016 \geq 0.35$.

If only the sensitivity rises from 0.80 to 1.80, then

$$D \approx 1.1665, \quad c \approx 0.2647,$$

so the same threshold causes abstention.

Mathematical check. The score decreases monotonically with both w and s because the two partial derivatives are nonpositive. The boundedness statement requires the stated condition $c_{ev} \in [0, 1]$ or explicit clipping. The numerical calculations give $D = 0.749833 \dots$, $c = 0.401578 \dots$, and, after raising s to 1.8, $c = 0.264737 \dots$

What this does not prove. The score c is not derived as a probability of correctness. Its exponential form is an engineering choice that gives smooth, bounded, monotone decay. The threshold τ controls a coverage-versus-risk tradeoff and must be selected using development/evaluation data rather than interpreted as a universal probability cutoff.

Equation (10): a stress test with proportional rather than constant bias

This equation is not the fitted measurement model

Equation (2) assumes a target-group shift that adds the same amount to each target row for a given feature. Equation (10) is deliberately different. It is used to generate test data where the shift is proportional to the clean value, so larger clean values move more than smaller ones.

This is a **stress test**: the estimator is asked to operate when the data-generating process violates one of its own simplifying assumptions.

New words used here. A **stress test** deliberately creates a difficult or mismatched situation to see how a method behaves. **Model mismatch** means the process that generated the test data differs from the process assumed by the fitted model.

Build the proportional multiplier

Let $a_j B$ be the intended signed average shift for feature j . It has the same physical unit as the feature. Let $\mu_{\text{tgt},j}$ be the clean mean of that feature in the target group.

The ratio

$$\frac{a_j B}{\mu_{\text{tgt},j}}$$

is unit-free because the numerator and denominator use the same unit. Call this ratio r_j . Then multiplying a clean value by $1 + r_j$ gives a proportional change:

$$y_{ij} = x_{ij}(1 + r_j).$$

Substituting the definition above and including the target-group switch gives

$$y_{ij} = x_{ij} \left[1 + \frac{a_j B}{\mu_{\text{tgt},j}} \mathbf{1}[g_i = \text{tgt}] \right]. \quad (10)$$

The indicator is zero for reference rows, leaving their values unchanged; it is one for target rows, which receive the proportional shift.

Thus Equation (10) states the target-only rule directly instead of using a separate piecewise condition.

In plain English. Equation (10) creates a different kind of distortion from Equation (2). Instead of adding the same number to every target row, it multiplies each target value by the same proportional factor.

Why the average shift still matches $a_j B$

For target rows, define

$$r_j = \frac{a_j B}{\mu_{\text{tgt},j}}.$$

Then

$$y_{ij} = x_{ij}(1 + r_j).$$

Average over all target rows:

$$\bar{y}_{\text{tgt},j} = \bar{x}_{\text{tgt},j}(1 + r_j).$$

If the denominator $\mu_{\text{tgt},j}$ is the same clean target mean, then

$$\bar{x}_{\text{tgt},j} = \mu_{\text{tgt},j}.$$

Therefore

$$\bar{y}_{\text{tgt},j} = \mu_{\text{tgt},j} \left(1 + \frac{a_j B}{\mu_{\text{tgt},j}} \right) = \mu_{\text{tgt},j} + a_j B.$$

Subtract the clean mean:

$$\bar{y}_{\text{tgt},j} - \bar{x}_{\text{tgt},j} = a_j B. \quad (10a)$$

So the *average* shift matches the additive benchmark even though individual row shifts are different.

Numerical example

Let the clean target values be

$$60, 75, 90.$$

Their mean is

$$\mu_{\text{tgt}} = 75.$$

Choose $aB = 20$. Then

$$1 + \frac{20}{75} = 1.266666 \dots$$

The generated target values are

$$60(1.266666 \dots) = 76,$$

$$75(1.266666 \dots) = 95,$$

$$90(1.266666 \dots) = 114.$$

The individual shifts are 16, 20, and 24, so they are not constant. But the new mean is

$$(76 + 95 + 114)/3 = 95,$$

which is exactly 20 above the clean mean of 75.

Mathematical check. When $\mu_{\text{tgt},j}$ equals the clean target mean used in the denominator, averaging Equation (10) gives an exact mean increase of $a_j B$. The numerical example changes the mean from 75 to 95 while producing row-specific shifts of 16, 20, and 24.

What this does not prove. The denominator must be nonzero. A multiplicative distortion may also be inappropriate for features whose scale includes zero crossings, negative values, or strict physical bounds. The test generator therefore uses declared feature bounds/clipping where needed.

Equation (11): a stress test with unusually frequent extreme errors

Why use a different noise distribution?

Equation (2) models ordinary row-level error with a normal distribution. A normal distribution gives very large deviations relatively little probability. Real data can sometimes contain more extreme observations than a normal model would predict.

Equation (11) therefore creates a stress test using a Student- t distribution with three degrees of freedom, written t_3 . It is symmetric like a normal distribution but has heavier tails, meaning large deviations occur more often.

New words used here. A distribution has **heavy tails** when values far from the center occur more frequently than they would under a normal bell curve. **Degrees of freedom** is a parameter that controls the shape of a Student- t distribution; with 3 degrees of freedom, the tails are much heavier than a normal distribution.

Start with the systematic target shift

As in the additive benchmark, target rows receive a signed systematic shift

$$a_j B \mathbf{1}[g_i = \text{tgt}].$$

Reference rows receive zero from this term.

Add heavy-tailed row noise

Let

$$T_{ij} \sim t_3.$$

A standard t random variable with $\nu > 2$ degrees of freedom has variance

$$\text{Var}(T) = \frac{\nu}{\nu - 2}.$$

For $\nu = 3$,

$$\text{Var}(T) = \frac{3}{3 - 2} = 3.$$

Therefore its standard deviation is $\sqrt{3}$. Divide by $\sqrt{3}$:

$$\text{Var}\left(\frac{T}{\sqrt{3}}\right) = \frac{1}{3} \text{Var}(T) = 1.$$

So $T/\sqrt{3}$ has unit variance.

Now multiply it by

$$0.5 \sigma_{\text{ref},j}^{\text{clean}},$$

where $\sigma_{\text{ref},j}^{\text{clean}}$ is the clean reference-group standard deviation used only to set the severity of this synthetic test. The resulting noise has standard deviation

$$0.5 \sigma_{\text{ref},j}^{\text{clean}}.$$

Putting the systematic shift and heavy-tailed noise together gives

$$y_{ij} = x_{ij} + a_j B\mathbf{1}[g_i = \text{tgt}] + 0.5 \sigma_{\text{ref},j}^{\text{clean}} \frac{T_{ij}}{\sqrt{3}}, \quad T_{ij} \sim t_3. \quad (11)$$

In plain English. Equation (11) keeps an additive target-group shift but replaces ordinary normal row noise with a distribution that produces extreme positive or negative errors more often. The division by $\sqrt{3}$ is there only to standardize the variance before the desired noise scale is applied.

Verify the noise scale

Because

$$\text{Var}\left(\frac{T}{\sqrt{3}}\right) = 1,$$

multiplying by $0.5 \sigma_{\text{ref}}^{\text{clean}}$ gives

$$\text{SD}\left(0.5 \sigma_{\text{ref}}^{\text{clean}} \frac{T}{\sqrt{3}}\right) = 0.5 \sigma_{\text{ref}}^{\text{clean}}. \quad (11a)$$

For t_3 , the variance is finite, but the fourth moment is not finite. In practical terms, quantities that depend strongly on squared extreme values, such as sample variance, can be much less stable than under normal noise.

Numerical example

Let

$$\sigma_{\text{ref}}^{\text{clean}} = 10, \quad x = 75, \quad aB = 20, \quad T = 2.4$$

for a target row. The heavy-tailed noise term is

$$0.5(10) \frac{2.4}{\sqrt{3}} \approx 6.9282.$$

Therefore

$$y = 75 + 20 + 6.9282 \approx 101.9282.$$

For a reference row with $x = 70$ and $T = -1.2$, the systematic target term is zero. The noise is

$$0.5(10) \frac{-1.2}{\sqrt{3}} \approx -3.4641,$$

so

$$y \approx 66.5359.$$

Mathematical check. For $T \sim t_3$, the variance is exactly 3, so $T/\sqrt{3}$ has variance 1. Multiplication by $0.5\sigma_{\text{ref}}^{\text{clean}}$ therefore gives variance $0.25(\sigma_{\text{ref}}^{\text{clean}})^2$ and standard deviation $0.5\sigma_{\text{ref}}^{\text{clean}}$. The two numerical examples evaluate to approximately 101.9282 and 66.5359.

What this does not prove. This is one specific synthetic failure mode, not proof of robustness to every possible process that creates unusually extreme observations, noise distribution that is uneven around its center, dependence pattern, or situation in which different groups have different noise levels. The clean reference standard deviation is used by the test generator to control severity; it is not secret information handed to the estimator during deployment.

Putting the equations together in one worked example

The equations are easier to remember when viewed as one chain rather than eleven isolated formulas.

Suppose the recorded data show a target-minus-reference carbon gap of about 9.83. Equation (2) says that a recorded gap can be the clean gap plus a systematic target shift, so the recorded 9.83 should not automatically be treated as the clean gap.

Suppose an audit and prior anchor are combined by Equation (8). With

$$\bar{g}_{\text{audit}} = 5.6, \quad g_{\text{prior}} = 5.0, \quad n = 20, \quad k = 12,$$

the resolved anchor is

$$g^{\star} = 5.375.$$

Equations (3)–(5) then give more weight to hidden clean states whose group gaps and adverse-direction behavior are more consistent with the declared penalties and anchor. Suppose the retained corrected states produce

$$\hat{x} = 5.30, \quad \hat{v} = 0.16.$$

With residual-bias cushion $\sigma_b = 0.75$, Equation (6) gives

$$I = [3.634, 6.966], \quad w = 3.332.$$

Now perturb the anchor by $\delta = 2$. Suppose the estimate changes from 5.30 to 6.10. Equation (7) gives

$$s = 0.80.$$

Finally use

$$c_{ev} = 0.85, \quad w_0 = 4, \quad s_0 = 1.2.$$

Equation (9) gives

$$c \approx 0.4016.$$

If the answer threshold is $\tau = 0.35$, the system answers. If the anchor perturbation instead caused the estimate to move to 7.10, then $s = 1.80$ and the same fusion rule gives

$$c \approx 0.2647,$$

which falls below the threshold and causes abstention.

In plain English. The chain is: model how recorded values may differ from clean values; represent several possible clean explanations; downweight explanations that violate the declared gap/bias conditions; summarize uncertainty; test anchor dependence; then answer only when the final selection score clears the threshold.

Glossary in ordinary language

This glossary is included for quick reference, but the document does not rely on it: each term is also explained where it first appears.

Term	Meaning here
Anchor	Outside information about the expected clean difference between the target and reference groups.
Bias	A systematic shift rather than a purely random fluctuation.
Conditional	Evaluated while treating some stated information as given or fixed.
Covariance	A measure of whether two varying quantities tend to move together.
Density	A curve describing where values of a continuous random quantity are concentrated.
Distribution	A mathematical description of how probability or weight is spread over possible values.
Expected value	The model-based long-run average of a random quantity.
Finite difference	The change in an output divided by, or compared across, a deliberate nonzero change in an input.
Generalized Bayes	Updating a distribution using a loss function rather than requiring all new information to be expressed through an ordinary likelihood.
Heavy-tailed	Having more probability far from the center than a normal bell curve.
Hidden/latent value	A quantity not directly observed but represented inside the model.
Indicator	A 0/1 mathematical switch that turns a term off or on.
KL divergence	A nonnegative measure of how much one probability distribution differs from another.
MCMC	A computational method that generates a sequence of samples intended to represent a complicated target distribution.
Metropolis–Hastings	An MCMC rule that accepts or rejects proposed moves using a probability ratio.
Normal distribution	The familiar symmetric bell-shaped probability model.
Pseudocount	Prior weight written in count-like units; it need not represent real observed records.
Sensitivity	How much an answer changes when an input or assumption is deliberately changed.
Shrinkage	Pulling a noisy estimate toward a more stable reference value.

Term	Meaning here
Standard deviation	The square root of variance; a measure of spread in the original unit.
Standardization	Dividing by an appropriate scale to obtain a unit-free quantity.
Variance	A measure of spread based on squared distance from the mean.

Mathematical verification ledger

The following statements were checked after the explanatory rewrite, independently of the prose simplification.

Eq.	What was checked	Result / condition
1	algebra, conditional mean/variance, normal density	correct under stated zero-mean and variance assumptions
2	indicator logic, group expectations, gap identity	correct; clean gap and bias remain non-identifiable without outside information
3	exponential reweighting, finite-state normalization, optimization form, Metropolis–Hastings acceptance formula	correct target form; exact computer sampling still depends on the sampler rules and whether the generated sequence has settled
4	dimensions, nonnegativity, Euclidean expansion, numerical value	correct for positive scales h_j
5	direction logic, positive-part operation, dimensions, numerical value	correct for positive scales and declared directions
6	full variance identity, independence reduction, 1.96 multiplier, endpoints and width	correct as a nominal normal-style construction under the stated assumptions
7	units, absolute finite change, normalized slope example	correct diagnostic definition
8	weighted-average weights, derivative, positive second derivative, numerical value	unique minimum of the stated squared-disagreement function; arithmetic correct
9	unit-free danger score, bounds, derivatives, both numerical cases	correct; $[0, 1]$ bound requires $c_{ev} \in [0, 1]$ or clipping
10	unit cancellation, target-only scope, sample-mean identity, numerical example	exact mean shift $a_j B$ when denominator is the same nonzero clean target mean
11	t_3 variance, $\sqrt{3}$ standardization, resulting noise SD, numerical examples	correct

References and provenance

- [R1] R. J. Carroll, D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu, *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd ed., Chapman & Hall/CRC, 2006.
- [R2] P. G. Bissiri, C. C. Holmes, and S. G. Walker, “A General Framework for Updating Belief Distributions,” *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130, 2016. doi:10.1111/rssb.12158.
- [R3] W. K. Hastings, “Monte Carlo Sampling Methods Using Markov Chains and Their Applications,” *Biometrika*, 57(1):97–109, 1970. doi:10.1093/biomet/57.1.97.
- [R4] NIST/SEMATECH, *e-Handbook of Statistical Methods*, standard normal distribution and Student- t distribution reference material.
- [R5] Y. Geifman and R. El-Yaniv, “SelectiveNet: A Deep Neural Network with an Integrated Reject Option,” *PMLR* 97:2151–2159, 2019.
- [R6] A. Kamath, R. Jia, and P. Liang, “Selective Question Answering under Domain Shift,” *ACL*, pp. 5684–5696, 2020.
- [R7] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner, “Rank-Normalization, Folding, and Localization: An Improved $\hat{R}$ for Assessing Convergence of MCMC,” *Bayesian Analysis*, 16(2):667–718, 2021.

Source artifacts used for equation identity and interpretation: the supplied ICTAI equation-verification supplement, the current manuscript equation numbering described in that supplement, and the earlier verification notes that distinguish literature templates from system-specific definitions and robustness generators.